

Causal Inference for Intervention & Service Evaluations

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Welcome

This is a practical handbook on causal inference methods for evaluating policy interventions and service changes within the NHS, with a focus on leveraging observational real-world data to generate robust evidence.

The handbook was written initially as part of an NHS England PhD Internship done jointly with the NHS North Central London Integrated Care Board and University College London, and will continue to be periodically updated with revisions following feedback from academics, data scientists, and public health practitioners.



1 Introduction

1.1 Background & Motivation for Causal Inference in Service Evaluations

1.2 A Brief History on Conceptualising Causality

1.3 Common Assumptions in Causal Inference Methodology

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1.5 New Unifying Guidance on Choosing Causal Inference Methods (Method Selection Tree)

1.6 Key Resources

1.7 Coverage of Causal Inference Methods in Key Texts

Part I

Key Concepts in Causal Inference

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2 Randomized Controlled Trials

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3 Directed Acyclic Graphs

TBC

4 Target Trial Emulation

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5 Defining Treatment Strategies

TBC

Part II

Quasi-Experimental Methods

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6 Difference-in-Differences

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7.1 Standard Interrupted Time Series Analysis

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11 Extensions of Quasi-Experimental Designs

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11.1 Negative Controls

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Part III

Adjustment-Based Methods

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12 Outcome Regression (Selection on Observables)

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13 Matching (Exact & Covariate)

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Part IV

Applying Causal Methods: Worked Examples in the NHS

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19 Case Study 1a: DOAC Head-to-Head Emulated Trial

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20 Case Study 1b: DOAC Policy Prescription Change Interrupted Time Series Analyses

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21 # Case Study 2: Ophthalmology Hub of Care Model Interrupted Time Series Analyses

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22 Practical Barriers to Implementation: Common Challenges in Causal Inference

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23 Conclusion

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24 About the Author & Acknowledgements

24.1 About Me

Hi, my name is Jamie and I'm an MRC funded PhD student at UCL, focusing on the application of causal inference methods within Epidemiology, Data Science, and Health Economics.

Before my PhD, I previously graduated with a BSc in Population Health Sciences (Data Science) from UCL, and an NIHR funded MSc in Epidemiology from the London School of Hygiene and Tropical Medicine.

Prior to my PhD, my interest in researching the application of causal inference methods in observational data stemmed from my varied experience working across multiple areas of public health research, including:

- Performing secondary analysis on a nutritional labelling RCT at the NIHR Obesity Policy Research Unit
- Supporting cancer trials as a Clinical Trials Assistant in the MRC Clinical Trials Unit
- Identifying barriers to STI testing at the NIHR Health Protection Research Unit in Blood Borne and Sexually Transmitted Infections
- Quantifying the effects of acute and long-term exposure of air pollutants on respiratory related hospital admissions and related outcomes at UCL
- Evaluating the risk of long-term adverse events in cancer survivors using electronic health record data at LSHTM

Having worked with both randomized trials and observational data, focusing my research on causal inference in real-world data felt like a natural next step.

If you have any questions do feel free to reach out to me via the following channels:
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24.2 Acknowledgements

This handbook was written during my PhD placement with NHS England and the North Central London Integrated Care Board as part of the NHS England Data science PhD internship programme, supervised by Jonathan Pearson (NHSE) and Emily Baldwin (NCL ICB).